MATHEMATICAL PHYSICS

SEMESTER 1, REPEAT

2018–2019

MP363
Quantum Mechanics I

Dr. M. Haque, Prof. D. A. Johnston and Dr. J.-I. Skullerud

Time allowed: 2 hours

Answer **ALL** questions

1. The set $\{|\phi_1\rangle, |\phi_2\rangle\}$ is an orthonormal basis set for a 2-dimensional Hilbert space. The states $|W\rangle$, $|\theta_1\rangle$ and $|\theta_2\rangle$ are defined as

$$|W\rangle = -i\alpha\,|\phi_1\rangle + \alpha\,|\phi_2\rangle\,,$$

$$|\theta_1\rangle = \frac{1}{\sqrt{2}}\,|\phi_1\rangle - \frac{i}{\sqrt{2}}\,|\phi_2\rangle\,, \qquad |\theta_2\rangle = \frac{i}{\sqrt{3}}\,|\phi_1\rangle + \frac{\sqrt{2}}{\sqrt{3}}\,|\phi_2\rangle\,.$$

(a) The state $|W\rangle$ is normalized. The real and imaginary parts of the constant α are equal. Find α.

[6 marks]

(b) Find out and explain whether or not the set $\{|\theta_1\rangle, |\theta_2\rangle\}$ is an orthonormal set.

[4 marks]

(c) Using the representation

$$|\phi_1\rangle = \begin{pmatrix}1\\0\end{pmatrix}\,, \qquad |\phi_2\rangle = \begin{pmatrix}0\\1\end{pmatrix}\,,$$

express the operator $\hat{M} = |\theta_1\rangle\langle\theta_2|$ as a matrix. Explain whether or not the operator $\hat{M}$ is hermitian.

[5 marks]

(d) The observable B is represented by the operator $\hat{B} = |\theta_1\rangle\langle\theta_1|$. Calculate the uncertainty of the observable B in the state $|\phi_1\rangle$.

[10 marks]

(e) If B is measured, what are the possible answers that can be obtained?

[8 marks]

2. Consider a particle of mass m in a one-dimensional harmonic oscillator potential $V(x) = \frac{1}{2}m\omega^2 x^2$. We denote the orthonormalized energy eigenstates of the harmonic oscillator by $|n\rangle$, on which the lowering and raising operators act as follows:

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle \qquad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

The ladder operators are defined as

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\frac{1}{\sigma}\hat{x} + \frac{i\sigma}{\hbar}\hat{p}\right), \qquad \hat{a}^\dagger = \frac{1}{\sqrt{2}}\left(\frac{1}{\sigma}\hat{x} - \frac{i\sigma}{\hbar}\hat{p}\right),$$

where $\sigma = \sqrt{\hbar/(m\omega)}$. Here $\hat{x}$ is the position operator and $\hat{p}$ is the momentum operator.

(a) Find the uncertainty of momentum in the state $|n\rangle$.

[13 marks]

(b) If the system wavefunction at time $t = 0$ is

$$|\psi(0)\rangle = \frac{i}{2}\,|2\rangle - \frac{\sqrt{3}}{2}\,|3\rangle$$

then what is the wavefunction at a later time $t = T$? Your answer should contain the oscillator frequency ω.

If the energy is measured at time $t = T$, what are the possible results of the measurement? What are the probabilities for the possible results to occur?

[7 marks]

(c) Given that the initial system wavefunction is $|\psi(0)\rangle = \frac{i}{2}\,|2\rangle - \frac{\sqrt{3}}{2}\,|3\rangle$, calculate the expectation value of position in the state $|\psi(t)\rangle$ as a function of time. Sketch a plot of $\langle\psi(t)|\,\hat{x}\,|\psi(t)\rangle$ against time.

[14 marks]

3. Consider a particle of mass m on an infinite one-dimensional line. It is subject to the 'step' potential

$$V(x) = \begin{cases} 0 & \text{for } x \leq 0 \qquad \text{(region 1)} \\ V_0 & \text{for } x > 0 \qquad \text{(region 2)} \end{cases}$$

where V_0 is a positive constant. For $E > V_0$, the time-independent Schroedinger equation has a solution with energy E, of the form

$$\begin{aligned} \psi(x) &= \psi_1(x) = e^{ik_1x} + Ae^{-ik_1x} && \text{(region 1)} \\ \psi(x) &= \psi_2(x) = Be^{ik_2x} && \text{(region 2)} \end{aligned}$$

where k_1 and k_2 are real and positive.

(a) Express the constants k_1 and k_2 in terms of E and V_0.

[5 marks]

(b) The solution written above can be interpreted as a scattering situation. Identify the terms representing incident, reflected and transmitted waves.

[3 marks]

(c) What are the boundary conditions that the solution must satisfy at $x = 0$?

Using these boundary conditions, calculate the constants A and B in terms of k_1 and k_2.

[8 marks]

(d) Define the reflection probability (R) and transmission probability (T) in terms of the constants appearing in the wavefunction.

Explain what you expect for R and T in the limit $E \gg V_0$.

[8 marks]

(e) Now consider the time-independent solution for $E < V_0$.

In region 2, the wavefunction is now of the form $\psi(x) = Ce^{\alpha x} + De^{-\alpha x}$, where α is real and positive.

Explain whether/why C or D needs to be zero.

Find A, C, D in terms of k_1 and α.

[9 marks]

_____*_____

PARTIAL ANSWERS

_____*_____

1. Question 1.

The set $\{|\phi_1\rangle, |\phi_2\rangle\}$ is an orthonormal basis set for a 2-dimensional Hilbert space. The states $|W\rangle$, $|\theta_1\rangle$ and $|\theta_2\rangle$ are defined as

$$|W\rangle = -i\alpha\,|\phi_1\rangle + \alpha\,|\phi_2\rangle\,,$$
$$|\theta_1\rangle = \frac{1}{\sqrt{2}}\,|\phi_1\rangle - \frac{i}{\sqrt{2}}\,|\phi_2\rangle\,, \qquad |\theta_2\rangle = \frac{i}{\sqrt{3}}\,|\phi_1\rangle + \frac{\sqrt{2}}{\sqrt{3}}\,|\phi_2\rangle\,.$$

(a) Question 1(a)

The state $|W\rangle$ is normalized. The real and imaginary parts of the constant α are equal. Find α.

[6 marks]

[Sample Answer:]

$$|-i\alpha|^2 + |\alpha|^2 = 1 \quad \Longrightarrow |\alpha|^2 = \frac{1}{2} \quad \Longrightarrow |\alpha| = \frac{1}{\sqrt{2}}$$

Since the real and imaginary parts of α are equal, we can write

$$\alpha = b + ib$$

where b is a real number. Then $|\alpha|^2 = b^2 + b^2 = 2b^2$. Hence

$$2b^2 = \frac{1}{2} \quad \Longrightarrow b = \pm\frac{1}{2}$$

Therefore we have

$$\text{either} \quad \alpha = \frac{1}{2} + \frac{i}{2} \quad \text{or} \quad \alpha = -\frac{1}{2} - \frac{i}{2}$$

-=-=-=-= * =-=-=-=-

(b) Question 1(b)

Find out and explain whether or not the set $\{|\theta_1\rangle, |\theta_2\rangle\}$ is an orthonormal set.

[4 marks]

[Sample Answer:]
To be an orthonormal set, the states have to be **normalized** and mutually **orthogonal**.
The two states $|\theta_1\rangle$ and $|\theta_2\rangle$ are each normalized.
Are they orthogonal to each other?

$$\begin{aligned}\langle\theta_1|\theta_2\rangle &= \left(\frac{1}{\sqrt{2}}\langle\phi_1| + \frac{i}{\sqrt{2}}\langle\phi_2|\right)\left(\frac{i}{\sqrt{3}}|\phi_1\rangle + \frac{\sqrt{2}}{\sqrt{3}}|\phi_2\rangle\right) \\ &= \frac{1}{\sqrt{2}}\cdot\frac{i}{\sqrt{3}} + \frac{i}{\sqrt{2}}\cdot\frac{\sqrt{2}}{\sqrt{3}} = \frac{1+\sqrt{2}}{\sqrt{6}}i \neq 0\end{aligned}$$

Not orthogonal to each other! Thus the set is not an orthonormal set.

-=-=-=-= * =-=-=-=-

(c) Question 1(c)

Using the representation

$$|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |\phi_2\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

express the operator $\hat{M} = |\theta_1\rangle\langle\theta_2|$ as a matrix. Explain whether or not the operator $\hat{M}$ is hermitian.

[5 marks]

[Sample Answer:]

$$|\theta_1\rangle = \frac{1}{\sqrt{2}}|\phi_1\rangle - \frac{i}{\sqrt{2}}|\phi_2\rangle = \begin{pmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix}$$

$$|\theta_2\rangle = \frac{i}{\sqrt{3}}|\phi_1\rangle + \frac{\sqrt{2}}{\sqrt{3}}|\phi_2\rangle = \begin{pmatrix} i/\sqrt{3} \\ \sqrt{2}/\sqrt{3} \end{pmatrix}$$

$$\hat{M} = |\theta_1\rangle\langle\theta_2| = \begin{pmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix} \begin{pmatrix} -i/\sqrt{3} & \sqrt{2}/\sqrt{3} \end{pmatrix} = \begin{pmatrix} -\frac{i}{\sqrt{6}} & +\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & -\frac{i}{\sqrt{3}} \end{pmatrix}$$

$$= \frac{1}{\sqrt{6}} \begin{pmatrix} -i & +\sqrt{2} \\ -1 & -i\sqrt{2} \end{pmatrix}$$

The matrix is not hermitian because

$$\hat{M}^\dagger = \frac{1}{\sqrt{6}} \begin{pmatrix} +i & -1 \\ +\sqrt{2} & +i\sqrt{2} \end{pmatrix} \neq \hat{M}$$

-=-=-=-= * =-=-=-=-

(d) Question 1(d)

The observable B is represented by the operator $\hat{B} = |\theta_1\rangle\langle\theta_1|$. Calculate the uncertainty of the observable B in the state $|\phi_1\rangle$.

[10 marks]

[Sample Answer:]

$$|\theta_1\rangle \;=\; \frac{1}{\sqrt{2}}\,|\phi_1\rangle \;-\; \frac{i}{\sqrt{2}}\,|\phi_2\rangle \;=\; \begin{pmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix}$$

$$\hat{B} \;=\; |\theta_1\rangle\langle\theta_1| \;=\; \begin{pmatrix} 1/\sqrt{2} \\ -i/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & +i/\sqrt{2} \end{pmatrix} \;=\; \begin{pmatrix} \frac{1}{2} & +\frac{i}{2} \\ -\frac{i}{2} & \frac{1}{2} \end{pmatrix}$$
$$= \; \frac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}$$

$$\hat{B}^2 \;=\; \frac{1}{4}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \;=\; \frac{1}{4}\begin{pmatrix} 2 & 2i \\ -2i & 2 \end{pmatrix} \;=\; \frac{1}{2}\begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} \;=\; \hat{B}$$

In the state $|\phi_1\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$,

$$\langle B\rangle \;=\; \langle\theta_1|\,\hat{B}\,|\theta_1\rangle \;=\; \begin{pmatrix} 1 & 0 \end{pmatrix}\hat{B}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \;=\; B_{11} \;=\; \frac{1}{2}$$

and

$$\langle B^2\rangle \;=\; \langle B\rangle \;=\; \frac{1}{2}$$

so that

$$\langle B^2\rangle - \langle B\rangle^2 \;=\; \frac{1}{2} - \frac{1}{4} \;=\; \frac{1}{4}$$

The uncertainty is

$$\Delta B \;=\; \sqrt{\langle B^2\rangle - \langle B\rangle^2} \;=\; \frac{1}{2}$$

-=-=-=-= * =-=-=-=-

(e) Question 1(e)
If B is measured, what are the possible answers that can be obtained?

[8 marks]

[Sample Answer:]
The eigenvalues of the B matrix are the possible results of a measurement. Calculation shows the eigenvalues to be 0 and 1. These are the possible values.
The eigenvalue calculation might go as follows:

$$\begin{vmatrix} \frac{1}{2}-\lambda & \frac{i}{2} \\ -\frac{i}{2} & \frac{1}{2}-\lambda \end{vmatrix} = 0 \qquad \Longrightarrow \qquad \left(\frac{1}{2}-\lambda\right)^2 - \frac{i}{2}\left(-\frac{i}{2}\right) = 0$$
$$\Longrightarrow \quad \frac{1}{4} + \lambda^2 - \lambda - \frac{1}{4} = 0$$
$$\Longrightarrow \quad \lambda^2 - \lambda = 0 \qquad \Longrightarrow \qquad \lambda = 0, 1$$

-=-=-=-= * =-=-=-=-

2. Question 2.

Consider a particle of mass m in a one-dimensional harmonic oscillator potential $V(x) = \frac{1}{2}m\omega^2x^2$. We denote the orthonormalized energy eigenstates of the harmonic oscillator by $|n\rangle$, on which the lowering and raising operators act as follows:

$$\hat{a}\,|n\rangle = \sqrt{n}\,|n-1\rangle \qquad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\,|n+1\rangle$$

The ladder operators are defined as

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\frac{1}{\sigma}\hat{x} + \frac{i\sigma}{\hbar}\hat{p}\right), \qquad \hat{a}^\dagger = \frac{1}{\sqrt{2}}\left(\frac{1}{\sigma}\hat{x} - \frac{i\sigma}{\hbar}\hat{p}\right),$$

where $\sigma = \sqrt{\hbar/(m\omega)}$. Here $\hat{x}$ is the position operator and $\hat{p}$ is the momentum operator.

(a) Question 2(a)
Find the uncertainty of momentum in the state $|n\rangle$.

[13 marks]

[Sample Answer:]
Inverting the given definitions,

$$\hat{x} = \frac{\sigma}{\sqrt{2}}\left(\hat{a} + \hat{a}^\dagger\right) \qquad \hat{p} = \frac{-i\hbar}{\sqrt{2}\sigma}\left(\hat{a} - \hat{a}^\dagger\right)$$

Thus

$$\hat{p}^2 \;=\; -\frac{\hbar^2}{2\sigma^2}\left(\hat{a} - \hat{a}^\dagger\right)\left(\hat{a} - \hat{a}^\dagger\right) \;=\; -\frac{\hbar^2}{2\sigma^2}\left(\hat{a}\hat{a} - \hat{a}\hat{a}^\dagger - \hat{a}^\dagger\hat{a} + \hat{a}^\dagger\hat{a}^\dagger\right) \quad (1)$$

NOTE!! Since $\hat{a}$ and $\hat{a}^\dagger$ do not commute, $\hat{a}^\dagger\hat{a} \neq \hat{a}\hat{a}^\dagger$. The ordering of operators matter; thus $-\hat{a}^\dagger\hat{a} - \hat{a}\hat{a}^\dagger \neq -2\hat{a}^\dagger\hat{a}$ This might be a common mistake.
To evaluate the uncertainty $\Delta p = \sqrt{\langle\hat{p}^2\rangle - \langle\hat{p}\rangle^2}$ in the state $|n\rangle$, we need to calculate $\langle\hat{p}\rangle = \langle n|\,\hat{p}\,|n\rangle$ and $\langle\hat{p}^2\rangle = \langle n|\,\hat{p}^2\,|n\rangle$. We first calculate the expectation values of the required ladder operator combinations:

$$\langle n|\,\hat{a}\,|n\rangle = \sqrt{n}\langle n\big|n-1\rangle = 0 \qquad \langle n|\,\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\langle n\big|n+1\rangle = 0$$

Note we are using the orthonormality of the eigenstates, $\langle m\big|n+1\rangle = \delta_{mn}$. Extending this calculation, one sees that $\hat{a}^\dagger\hat{a}^\dagger$ and $\hat{a}\hat{a}$ have zero expectation value, but $\hat{a}^\dagger\hat{a}$ and $\hat{a}\hat{a}^\dagger$ give nonzero contributions.

$$\langle n|\,\hat{a}\hat{a}\,|n\rangle = \sqrt{n}\langle n|\,\hat{a}\,|n-1\rangle = \sqrt{n}\sqrt{n-1}\langle n\big|n-2\rangle = 0$$
$$\text{Similarly,} \qquad \langle n|\,\hat{a}^\dagger\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\sqrt{n+2}\langle n\big|n+2\rangle = 0$$
$$\langle n|\,\hat{a}^\dagger\hat{a}\,|n\rangle = \sqrt{n}\langle n|\,\hat{a}^\dagger\,|n-1\rangle = \sqrt{n}\sqrt{n}\langle n\big|n\rangle = n \times 1 = n$$
$$\langle n|\,\hat{a}\hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\langle n|\,\hat{a}\,|n+1\rangle = \sqrt{n+1}\sqrt{n+1}\langle n\big|n\rangle = n+1$$

Employing these, we obtain

$$\langle\hat{p}\rangle \;=\; \frac{-i\hbar}{\sqrt{2}\sigma}\left(\langle\hat{a}\rangle - \langle\hat{a}^\dagger\rangle\right) \;=\; 0$$

$$\langle\hat{p}^2\rangle \;=\; -\frac{\hbar^2}{2\sigma^2}\left(\langle\hat{a}\hat{a}\rangle - \langle\hat{a}\hat{a}\rangle^\dagger - \langle\hat{a}^\dagger\hat{a}\rangle + \langle\hat{a}^\dagger\hat{a}^\dagger\rangle\right) \;=\; -\frac{\hbar^2}{2\sigma^2}\Big[0 - (n+1) - n + 0\Big]$$

$$\Longrightarrow \qquad \langle\hat{p}^2\rangle \;=\; \frac{\hbar^2}{\sigma^2}\left(n + \frac{1}{2}\right)$$

Thus the unceratinty is

$$\Delta p = \sqrt{\langle \hat{p}^2 \rangle - \langle \hat{p} \rangle^2} = \frac{\hbar}{\sigma}\sqrt{n + \frac{1}{2}}$$

-=-=-=-= * =-=-=-=-

(b) Question 2(b)
If the system wavefunction at time $t = 0$ is

$$|\psi(0)\rangle \;=\; \frac{i}{2}\,|2\rangle - \frac{\sqrt{3}}{2}\,|3\rangle$$

then what is the wavefunction at a later time $t = T$? Your answer should contain the oscillator frequency ω.
If the energy is measured at time $t = T$, what are the possible results of the measurement? What are the probabilities for the possible results to occur?

[7 marks]

[Sample Answer:]
The two eigenstates appearing in the wavefunction correspond to eigenenergies

$$\epsilon_2 = \frac{5}{2}\hbar\omega\,; \quad \epsilon_3 = \frac{7}{2}\hbar\omega\,.$$

Hence the wavefunction at time $t = T$ is

$$\begin{aligned} |\psi(T)\rangle \;&=\; \frac{i}{2}\,|2\rangle e^{-i\epsilon_2 T/\hbar} - \frac{\sqrt{3}}{2}\,|3\rangle e^{-i\epsilon_3 T/\hbar} \\ &=\; \frac{i}{2}\,|2\rangle e^{-i5\omega T/2} - \frac{\sqrt{3}}{2}\,|3\rangle e^{-i7\omega T/2} \end{aligned}$$

The possible results of an energy measurement are the eigenenergies corresponding to the two eigenstates appearing in the wavefunction:

$$\epsilon_2 = \frac{5}{2}\hbar\omega\,; \quad \epsilon_3 = \frac{7}{2}\hbar\omega\,.$$

These appear with the probabilities

$$\left|\frac{i}{2}e^{-i5\omega T/2}\right|^2 \;=\; \frac{1}{4}\,, \quad \left|-\frac{\sqrt{3}}{2}e^{-i7\omega T/2}\right|^2 \;=\; \frac{3}{4}\,,$$

respectively. Note this is independent of T, as the phase factor (containing the time dependence) has modulus one and hence contributes nothing to the probability. It doesn't actually matter at which time the measurement is performed,

-=-=-=-= * =-=-=-=-

(c) Question 2(c)

Given that the initial system wavefunction is $|\psi(0)\rangle \;=\; \frac{i}{2}\,|2\rangle - \frac{\sqrt{3}}{2}\,|3\rangle$, calculate the expectation value of position in the state $|\psi(t)\rangle$ as a function of time. Sketch a plot of $\langle\psi(t)|\,\hat{x}\,|\psi(t)\rangle$ against time.

[14 marks]

[Sample Answer:]
The state at time t is

$$|\psi(t)\rangle \;=\; \frac{i}{2}\,|2\rangle e^{-i5\omega t/2} - \frac{\sqrt{3}}{2}\,|3\rangle e^{-i7\omega t/2}$$

Therefore

$$\begin{aligned}\langle\hat{x}\rangle \;&=\; \langle\psi(t)|\,\hat{x}\,|\psi(t)\rangle \\ =\; &\left(\frac{-i}{2}\langle 2|\,e^{+i5\omega t/2} - \frac{\sqrt{3}}{2}\langle 3|\,e^{+i7\omega t/2}\right)\hat{x}\left(\frac{i}{2}\,|2\rangle e^{-i5\omega t/2} - \frac{\sqrt{3}}{2}\,|3\rangle e^{-i7\omega t/2}\right)\end{aligned}$$

with

$$\hat{x} = \frac{\sigma}{\sqrt{2}}\left(\hat{a} + \hat{a}^\dagger\right)$$

In the expression for $\langle\hat{x}\rangle$ we have the following eight matrix elements appearing:

$$\begin{array}{cccc} \langle 2|\,\hat{a}\,|2\rangle & \langle 2|\,\hat{a}\,|3\rangle & \langle 3|\,\hat{a}\,|2\rangle & \langle 3|\,\hat{a}\,|3\rangle \\ \langle 2|\,\hat{a}^\dagger\,|2\rangle & \langle 2|\,\hat{a}^\dagger\,|3\rangle & \langle 3|\,\hat{a}^\dagger\,|2\rangle & \langle 3|\,\hat{a}^\dagger\,|3\rangle \end{array}$$

Of these, only the following two are nonzero:

$$\langle 2|\,\hat{a}\,|3\rangle = \sqrt{3}\langle 2|2\rangle \;=\; \sqrt{3}\,, \qquad \langle 3|\,\hat{a}^\dagger\,|2\rangle = \sqrt{3}\langle 3|3\rangle \;=\; \sqrt{3}\,,$$

while the other matrix elements are all zero. Thus

$$\begin{aligned}
\langle \hat{x} \rangle &= \frac{\sigma}{\sqrt{2}} \left(\left[-\frac{i}{2} \right] e^{+i5\omega t/2} \langle 2 | \, \hat{a} \, | 3 \rangle \left[-\frac{\sqrt{3}}{2} \right] e^{-i7\omega t/2} \right. \\
&\qquad \left. + \left[-\frac{\sqrt{3}}{2} \right] e^{+i7\omega t/2} \langle 3 | \, \hat{a}^\dagger \, | 2 \rangle \frac{i}{2} e^{-i5\omega t/2} \right) \\
&= \frac{\sigma}{\sqrt{2}} \frac{3i}{4} \left(-e^{-i\omega t} + e^{i\omega t} \right) \;=\; \frac{\sigma}{\sqrt{2}} \frac{3i}{4} \times 2i \sin(\omega t) \\
&= -\frac{3\sigma}{2\sqrt{2}} \sin(\omega t)
\end{aligned}$$

The required plot of $\langle \hat{x} \rangle$ versus time is a plot of the negative sine function.

-=-=-=-= * =-=-=-=-

3. Question 3.

Consider a particle of mass m on an infinite one-dimensional line. It is subject to the 'step' potential

$$V(x) = \begin{cases} 0 & \text{for } x \leq 0 \qquad \text{(region 1)} \\ V_0 & \text{for } x > 0 \qquad \text{(region 2)} \end{cases}$$

where V_0 is a positive constant. For $E > V_0$, the time-independent Schroedinger equation has a solution with energy E, of the form

$$\begin{aligned}
\psi(x) &= \psi_1(x) = e^{ik_1 x} + A e^{-ik_1 x} \qquad &\text{(region 1)} \\
\psi(x) &= \psi_2(x) = B e^{ik_2 x} \qquad &\text{(region 2)}
\end{aligned}$$

where k_1 and k_2 are real and positive.

(a) Question 3(a)

Express the constants k_1 and k_2 in terms of E and V_0.

[5 marks]

[Sample Answer:]

$$k_1 = \sqrt{\frac{2mE}{\hbar^2}}\,; \qquad k_2 = \sqrt{\frac{2m(E - V_0)}{\hbar^2}}$$

-=-=-=-= * =-=-=-=-

(b) Question 3(b)

The solution written above can be interpreted as a scattering situation. Identify the terms representing incident, reflected and transmitted waves.

[3 marks]

[Sample Answer:]

$e^{ik_1x} \longrightarrow$ incident wave

$Ae^{-ik_1x} \longrightarrow$ reflected wave

$Be^{ik_2x} \longrightarrow$ transmitted wave

-=-=-=-= * =-=-=-=-

(c) Question 3(c)

What are the boundary conditions that the solution must satisfy at $x = 0$?

Using these boundary conditions, calculate the constants A and B in terms of k_1 and k_2.

[8 marks]

[Sample Answer:]

Boundary conditions: the wavefunction and its derivative must be continuous at $x = 0$. Thus four boundary conditions:

$$\psi_1(0) = \psi_2(0)\,; \quad \psi_1'(0) = \psi_2'(0)$$

Each gives a relation involving A, B, k_1, k_2. The two equations are

$$1 + A = B$$
$$ik_1 - ik_1A = ik_2B$$

This is a system of two linear equations for the two constants A and B. Solving for A and B gives

$$A = \frac{k_1 - k_2}{k_1 + k_2}\,, \qquad B = \frac{2k_1}{k_1 + k_2}$$

-=-=-=-= * =-=-=-=-

(d) Question 3(d)

Define the reflection probability (R) and transmission probability (T) in terms of the constants appearing in the wavefunction.

Explain what you expect for R and T in the limit $E \gg V_0$.

[8 marks]

[**Sample Answer:**]

The reflection probability is the reflected current divided by the incident current.

The transmission probability is the transmitted current divided by the incident current.

$$\begin{aligned} \text{incident current} &= 1 \times \frac{\hbar k_1}{m} \\ \text{reflected current} &= |A|^2 \times \frac{\hbar k_1}{m} \\ \text{transmitted current} &= |B|^2 \times \frac{\hbar k_2}{m} \end{aligned}$$

Thus

$$R = |A|^2 \qquad T = |B|^2 \frac{k_2}{k_1}$$

For $E \gg V_0$, the barrier is negligible, we thus expect total transmission and no reflection, i.e., $R = 0$ and $T = 1$.

-=-=-=-= * =-=-=-=-

(e) Question 3(e)

Now consider the time-independent solution for $E < V_0$.

In region 2, the wavefunction is now of the form $\psi(x) = Ce^{\alpha x} + De^{-\alpha x}$, where α is real and positive.

Explain whether/why C or D needs to be zero.

Find A, C, D in terms of k_1 and α.

[9 marks]

[Sample Answer:]

Region 2 runs from $x = 0$ to $x = \infty$. In this region, $Ce^{\alpha x}$ is unbounded for any nonzero C and positive α. Hence, for the wavefunction to be make sense, we require $C = 0$.

To calculate A and D, we use the boundary conditions $\psi_1(0) = \psi_2(0)$ and $\psi_1'(0) = \psi_2'(0)$. These give:

$$1 + A = D \,, \qquad ik_1 - ik_1 A = -\alpha D$$

Solving for A and D gives

$$A \;=\; \frac{-\alpha - ik_1}{\alpha - ik_1} \,, \qquad D \;=\; \frac{-2ik_1}{\alpha - ik_1}$$

-=-=-=-= * =-=-=-=-